

IET, DAVV, Indore
B.E. II(ETC)
Class Test III (Nov 2023)
3ATRC1: Applied Mathematics-III

Time: 70 min

Max Marks: 20

Note: Attempt any FOUR questions. All questions carry equal marks.

Q.1. Two telecom companies, Provider A and Provider B, are competing for subscribers. Provider A's customer exhibit a 85% loyalty rate, while 60% of Provider B's former customers switch back to Provider A's service. (05)

- Develop a state transition matrix illustrating customer movement between the two providers.
- Determine the probability that a customer will be using Provider A's service after two periods if they initially started with provider B.

Q.2. Test whether the following matrices are regular stochastic matrices: (05)

$$A = \begin{bmatrix} 0 & 1 \\ 1/2 & 1/2 \end{bmatrix} \quad , \quad B = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$$